

Does the legacy of the American frontier make people today less supportive of environmental regulation and collective action to address climate change?

Research Module Macroeconomics
Economic Growth and Gender Dynamics in Historical Perspective

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1 Introduction

The American historian Frederick Jackson Turner famously argued that the experience of the American westward-expansion and colonization of land in the 18th and 19th century, the so called "American frontier", still shapes today's beliefs about politics and society (Turner, 1893). Although this theory remains controversial¹, Bazzi et al. (2020) provide a convincing data analysis that proves Turner's thesis. The authors create a new measurement to express the "Total Frontier Experience" (TFE) in years and regress this on different cultural indicators for "rugged individualism" such as the frequency of uncommon names, the preference for the republican party and lower tax rates. They find that more years on the frontier actually is a robust indicator for more individualistic political beliefs today.

We want to extend this analysis by asking whether this frontier experience also shapes beliefs about climate policy today. We differentiate between climate policy opinions on two different types of policy strategies, namely top-down environmental regulation that is passed by the president or the congress and bottom-up collective action that is initiated by regular citizens and local authorities. There are many studies about how political ideology, culture and demography affects opinions on climate change and associated policies. For example we know that conservative white males are more likely to be climate change deniers (McCright and Dunlap, 2011). Regarding the political parties of the US, it is also known that there is a polarization between Democrats and Republicans when it comes to the perception of climate change (Guber, 2013; McCright et al., 2014). Hornsey et al. (2016) conduct a meta-analysis that shows how political ideology is among the strongest predictors in regard to beliefs in climate change, overshadowing education and experience of extreme weather events. When it comes to the motivations behind conservative climate skepticism, Campbell and Kay (2014) show that conservatives do not generally tend to be more climate skeptic because they distrust climate scientists, but because their proposed solutions do not align with their values. Similarly, Boven et al. (2018) argue that Republicans do not disagree about the existence of climate change but that their opposition towards climate change policies simply reflects identity-based and ideological differences with Democrats, who generally are more in favor of climate change policies. Building on this rich literature linking ideology and climate attitudes, we add a historical and cultural dimension by combining these findings with Turner's frontier experience thesis of "rugged individualism". Further studying this possible connection might help in understanding the opposition towards climate policies in some US-regions and the underlying mechanisms and historical roots of climate change scepticism and therefore improve US climate policy design.

We answer our research question by regressing the TFE measurement by Bazzi et al. (2020) on various estimated survey questions by the Yale Climate Opinion Map from the year 2020 on the US-county-level. We distinguish between the two policy approaches (top-down environmental regulation / bottom-up collective action) by using two different sets of questions per approach as the dependent variable, which leads to four questions in total.

¹Commenting on the ethnocentric, colonial and sexist implications of Turner's thesis is out of the scope of this term paper but should not stay unmentioned.

We use a mediation analysis and a moderation analysis and try to analyze the link between the TFE and republican political ideology in regard to beliefs about climate change. While the mediation analysis is used as another control mechanism to separate the effect of TFE and republican vote share on climate policy opinions, the moderation analysis should give us a potential explanation for the interaction between conservative ideology and the "stickiness" of frontier culture.

2 Background

The westward expansion of the United States in the 19th century lead to the colonization of land that was previously not governed by a centralized authority and lacked meaningful infrastructure. These regions were marked by low population density and remoteness from more populated areas. According to Turner (1893), these two features cultivated a strong form of individualism, i.e. a focus on self-reliance and independence, that still influences the regions culture today and lead to a strong aversion against any form of intervention by the government. This form of "rugged individualism", a term coined by republican president Herbert Hoover, is still used by academics and the public to explain differences in political preferences between americans and their european origins. In comparison to simple individualism, "rugged individualism" is a extreme form of culturally specific US-individualism that, as Alesina et al. (2001) put it, consists of "strengthened individualistic feelings and beliefs in equality of opportunity rather than equality of outcomes"².

2.1 Bazzi et al. (2020)

Bazzi et al. (2020) try to provide empirical evidence for this theory by using econometric methods. To this end, they create a new measurement for expressing the exposure to frontier culture quantitatively on the county-level, the Total Frontier Exposure (TFE). This new key figure shows how many years each county was a part of the American frontier between the years 1790 and 1890 and is calculated by using both population density and closeness to the frontier line as indicators³. The authors verify these indicators by showing that regions with low population densities showed specific demographic and cultural traits: their inhabitants were disproportionately young, male, foreign-born, illiterate and more individualistic, which is approximated by having infrequent names. This way, they show that frontier societies indeed had specific cultural traits and were inhabited by more individualistic people in the past.

The study goes on to show that the frontier experience still shapes the affected counties to this day. 100 years later, higher TFE still correlates with more infrequent names today, which is again used as a proxy for individualism. The authors also show that higher TFE correlates with fiscal conservatism and opposition to redistribution. In practice, this leads to higher voting shares for the republican party and lower property tax rates. This can be seen as evidence for a persistent

²The authors define what they call "American anti-statism", which is also a extreme, U.S.-specific form of individualism, developed because of the frontier experience.

³See Section "Data" for a more precise definition of the TFE.

culture of "rugged individualism" as a result of frontier experience⁴.

At the end, Bazzi et al. (2020) describe the roots of frontier culture. There are two central mechanisms that explain the individualism on the frontier line. The first is selective migration: people that moved to the frontier were already more individualistic than the rest and people that moved away from the frontier into more settled counties were less individualistic than the rest. The second is cultural change: inhabitants that spend more time at the frontier became even more individualistic than before.

2.2 Hypotheses

One could argue that the frontier experience can therefore also influence these regions attitudes towards climate change as environmental policy in general requires coordination among individuals to protect public goods, a form of cooperation that tends to get in conflict with strong individualistic values. Bazzi et al. (2020) already show in their research that a higher TFE leads to stronger opposition towards the regulation of CO₂ emissions. We want to build on these findings by studying how exactly this persistence of "rugged individualism" shapes the opposition towards different forms of climate policy today. We therefore distinguish between two forms of policies that can both be used to combat climate change:

Environmental regulation: We define environmental regulation as laws and taxes that are enforced by a centralized government. In the US this would include the president, congress, the senate and governors. We call this approach "top-down" as policies are enforced by a higher political authority.

Collective action: In contrast to that, collective action is a more decentralized and individualistic approach towards climate change. Instead of centralized governmental choices, actions are coordinated and taken by citizens and local officials⁵. In contrast to top-down policies this approach is described as "bottom-up".

In theory, we should first expect to see the same result as Bazzi et al. (2020), namely that counties with higher TFE experience higher opposition to climate policy⁶. Because people living in counties with higher TFE are more individualistic, independent and self-reliant, they generally oppose both kinds of policy approaches. They are sceptical about collective problem-solving, whether as a top-down or a bottom-up approach, especially when the problem concerns a public good such as climate protection. Nevertheless, we should secondly expect a stronger aversion towards top-down environmental regulation than towards bottom-up collective action as a greater TFE leads to greater opposition towards big government than to local solutions. The latter, even though still

⁴The authors use different econometric approaches to prove the robustness of their findings, such as additional control variables and an instrumental variable.

⁵We classify local officials as bottom-up because decision-making remains locally embedded and closer to citizens.

⁶To be more precise, Bazzi et al. (2020) show that higher TFE leads to higher opposition towards the regulation of CO₂ emissions, like stated above.

disliked overall, may be less disliked. Bottom-up collective action in general is more compatible with "equality of opportunities" in contrast to "equality of outcomes" which may be closer to the top-down environmental regulation approach because coordination between regular citizens and local authorities is a more tolerable form of political action for individualistic people. Thirdly, counties with a higher republican vote share should experience a higher effect of the TFE than counties with a higher democratic vote share. Because the former are generally more conservative in their political ideology, they should tend to preserve their cultural habits. This should also include those habits formed by the frontier experience which are associated with "rugged individualism". So all in all, we formulate three hypotheses:

1. Higher TFE leads to higher opposition towards climate policies in general.
2. Higher TFE leads to greater opposition towards environmental regulation than towards collective action.
3. Republican counties experience a stronger effect of TFE.

3 Data

Our empirical analysis combines historical measures of frontier exposure with contemporary county-level data on climate change attitudes in the United States. All datasets are harmonized to consistent county boundaries to ensure comparability across sources and over time. After merging all sources and accounting for missing observations, the final sample consists of 2,036 U.S. counties.

3.1 Dependent Variable: YCOM

Table 1: Detailed Summary Statistics: YCOM Opinion Variables

Variable	Mean	Std. Dev.	Min	p25	p50	p75	Max
Citizens: Do Less	13.13	2.01	7	12	13	15	19
Local Officials: Do Less	16.98	2.27	9	16	17	19	23
Oppose High Priority	28.81	5.03	11	26	29	32	41
Oppose Carbon Tax	38.23	5.09	20	35	39	42	50

Note: All values are expressed in percentage points (share of county residents).

All outcome variables are taken from the Yale Climate Opinion Maps 2024 (YCOM) dataset. This dataset provides cross-sectional modeled public climate opinion estimates at the local-level across the United States. Because reliable survey data on climate attitudes rarely exist at sub-national scales, Howe et al. (2015) use multilevel regression and poststratification (MRP) to downscale national survey data and generate estimates for states, congressional districts, metropolitan areas, and counties.

MRP combines data from multiple national surveys and incorporates demographic, geographic, and time variables to partially pool information about respondents across subgroups, allowing for reliable estimation even in areas with sparse survey coverage. Building on the framework established by Howe et al. (2015), the 2024 version of YCOM used in this study employs data from 31 nationally representative surveys conducted from November 2008 through December 2024 ($n > 35,000$). The MRP process involves two steps. First, an initial multilevel regression step, individual survey responses⁷ are modelled as a function of both individual-level demographics⁸ and geography-level covariates⁹.

Second, during the poststratification stage, the multilevel regression results are used to project the average opinion of every demographic-geographic-year combination in the data. For example, the model can estimate the average belief of a Hispanic/Latina woman with a bachelor’s degree or higher living in Orange County, California in 2013. With 32 unique sex–race–education categories interacting with 3,143 counties, this generates 100,576 unique population types for county-level models. The final estimate for each geographic subunit is then obtained by weighting each population type’s projected belief by its true population count.

Our analysis uses four public climate change opinion variables from the YCOM dataset at the county level for the year 2020. Two capturing the top down environmental regulation approach and two other capturing the bottom up collective action approach. Each of the outcome variables is defined as the estimated share of adults in area i who agree with that item and is therefore expressed on a 0-100 percentage scale.

The following two public climate change opinion variables are chosen to represent environmental regulations:

1. Estimated percentage who do not say global warming should be a high priority for the next president and Congress
2. Estimated percentage who oppose requiring fossil fuel companies to pay a carbon tax and use the money to reduce other taxes (such as income tax) by an equal amount

These allow us to test whether frontier-shaped rugged individualism implies stronger opposition to centralized and redistributive climate policy.

The following two public climate change opinion variables are chosen to represent collective action:

3. Estimated percentage who think citizens should do less to address global warming
4. Estimated percentage who think their local officials should do less to address global warming

Allows testing whether frontier-shaped rugged individualism implies broader skepticism toward collective problem-solving, beyond opposition to federal intervention.

⁷For example, beliefs, attitudes, policy support, and so on.

⁸Gender, race, education, and a three-way interaction term among these variables.

⁹Percent of people who drive alone to work, percent of same-sex households, percent of CO2 emissions per capita, and percent of people who voted Democrat in the most recent election (2020).

For each of the four YCOM climate attitude questions, we select the response option capturing opposition to climate action, as this most directly reflects our theoretical interest in how Total Frontier Experience amplifies resistance to climate policy.

In separate regressions, each YCOM measure serves as the dependent variable, enabling us to examine how TFE relates to distinct dimensions of climate attitudes.

3.2 Main Regressor: TFE

Table 2: Detailed Summary Statistics: Total Frontier Experience in Years (TFE > 0)

Variable	Mean	Std. Dev.	Min	p25	p50	p75	Max
Total Frontier Experience	18.53	11.02	1	11	16	24	63

Note: p25, p50 (median), and p75 represent the interquartile range.

Our main explanatory variable is Total Frontier Experience (TFE), developed by Bazzi et al. (2020). The TFE measures the cumulative number of years a county was exposed to frontier conditions during the period of U.S. westward expansion, between 1790 and 1890. Counties with zero years of frontier exposure are excluded from the sample, and we also excluded counties in the west-coast frontier from the sample for this analysis. Following the definition proposed by Turner (1893) and formalized in the U.S. Census Bureau’s 1890 report, we define the frontier line as the boundary at which population density falls below two people per square mile (Bazzi et al., 2020). The frontier is thus composed of counties characterized by low population density and located in close geographic proximity to this boundary. Based on this time-varying classification, TFE is constructed as the cumulative number of years a county spent under frontier conditions between 1790 and 1890. Consequently, counties that were settled later or remained sparsely populated for longer periods exhibit higher TFE values.

Turner (1893) conceptualized frontier as a "form of society" rather than an area, by identifying characteristics such as isolation and abundance of resources. The TFE is designed to capture prolonged exposure to frontier conditions characterized by low population density, weak state capacity, limited infrastructure, and high reliance on individual self-sufficiency. A county is classified as frontier if its population density is below six people per square mile and it lies within 100 kilometers of the frontier line, where the frontier line itself is defined as the contour at two people per square mile. These thresholds are intended to capture frontier conditions that are consistent with Turner’s notion of the frontier as a distinct social environment.

To measure the duration of frontier exposure for each county, we follow the methodology developed by Bazzi et al. (2020). Specifically, they calculate the number of years each county spent in the frontier belt between 1790 and 1890 using a binary indicator for each year that denotes whether the county met the frontier criteria defined by proximity and low density. The Total Frontier Experience (TFE) for a county is then defined as the sum of these annual indicators. This approach allows us to quantify historical frontier exposure, capturing variation in the intensity and duration of frontier conditions across counties.

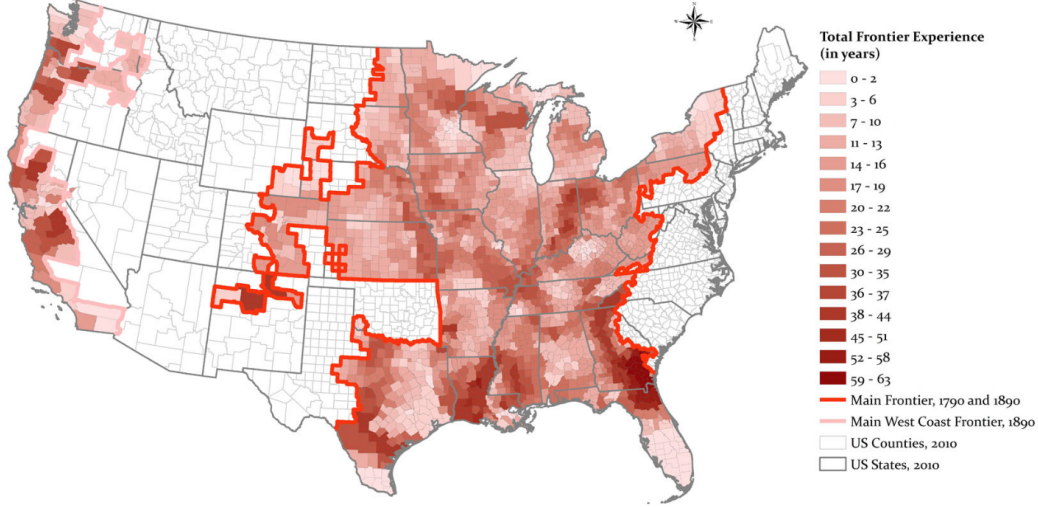


Figure 1: Map of TFE across U.S.-counties, *Source: Bazzi et al. (2020)*

3.3 Controls

Following Bazzi et al. (2020), our regressions include Geographic/Agroclimatic controls to absorb fixed spatial confounders. Characteristics, such as, county area, county centroid latitude and longitude, distance to oceans, lakes, rivers and portage sites from county centroid, mean county temperature and rainfall, elevation, and average potential agricultural yield and ruggedness, are largely time-invariant and shape long-run economic activities, settlement patterns, and exposure to environmental risk. Mirroring Bazzi et al. (2020), we also augment our baseline specification with a rich set of historical county characteristics, including the 1890 sex ratio, birthplace diversity, manufacturing employment share, immigrant share, years connected to the railroad by 1890, the 1860 slave population share, the 1890 Scottish–Irish immigrant share, and distance to the nearest mineral discovery site and Indian battle.

We control for contemporary county-level population density, measured as the logarithm of 2020 population density (persons per square mile) from the USDA Rural Atlas. Since frontier regions were historically defined by low density and modern rural counties differ systematically in political ideology and climate attitudes, including population density isolates the historical frontier effect from present-day settlement patterns and contemporary rural–urban differences.

Taken together these controls capture persistent demographic, economic, and cultural features that may both correlate with Total frontier exposure and independently shape contemporary climate attitudes.

3.4 Voting data

The political orientation of each county is estimated by the Republican vote share in the Presidential election. We use county-level presidential election returns from the MIT Election Data and Science Lab (MEDSL), to obtain U.S. presidential election returns in 2020 at county level.

Table 3: Detailed Summary Statistics: Republican Vote Share (Sample: TFE > 0)

Variable	Mean	Std. Dev.	Min	p25	p50	p75	Max
Rep. Vote Share (2020)	0.65	0.16	0.08	0.55	0.68	0.78	0.96

Note: U.S. Republican vote share represents the 2020 Presidential Election county total.

The 2020 raw data is isolated and used to calculate Republican vote share for 2020. The measure obtained is then merged with other covariates at county-level for further analysis. The variable is used in three ways: control variable for contemporary partisanship, diagnostic mediator in decomposition exercises, and interaction term in moderation analyses.

4 Empirical Strategy

We estimate the relationship between historical frontier exposure and today’s climate-policy attitudes, using four county-level outcomes from the Yale Climate Opinion Maps (YCOM) to capture climate-policy opposition.

To do this, we proceed in three steps. First, we assess whether TFE is associated with today’s climate-policy attitudes by regressing the estimated percentage expressing opposition to each YCOM question. Second, we replicate the central relationship emphasized in Bazzi et al. (2020) by estimating the effect of Total Frontier Experience (TFE) on Republican vote share. Next, motivated by the strong role of political ideology in climate attitudes, we explore whether the relationship between TFE and climate attitude operates through contemporary political ideology using diagnostic mediation analysis decomposition and, lastly, whether there is stronger effect of TFE in more republican counties using a moderation analysis with an interaction term.

Let i index counties and $s(i)$ denote the state of county i . All regressions are estimated at the county level and include the full set of geographic controls G_i , historical controls H_i , and state fixed effects $\delta_{s(i)}$. Standard errors are clustered at the 60km grid-cell level ¹⁰. To ensure consistent interpretation across specifications, we center our variables, y_i , TFE_i and R_i , around their sample mean so that the main effects are evaluated at average values of the variable.

4.1 Step 1: Frontier exposure and climate-policy opposition

We start by estimating whether frontier exposure is associated with climate-policy opposition today to address hypothesis 1 (higher TFE leading to higher opposition towards climate policies in general). Let y_i denote a county-level YCOM outcome (in percentage points, 0–100) for one of the four opposing questions, let TFE_i denote county i ’s total frontier experience in years, and let R_i denote county i ’s Republican vote share in the 2020 presidential election (scaled 0–1). Our

¹⁰This is important because counties in geographic proximity may share unobserved regional determinants of climate attitudes (e.g. political culture, economic structure, or regional media). Such correlation could otherwise understate standard errors and overstate statistical significance.

baseline regression is:

$$y_i = \beta_0 + \beta_1 TFE_i + G'_i \beta_2 + H'_i \beta_3 + \delta_{s(i)} + \varepsilon_i. \quad (1)$$

We estimate equation (1) for each of the four outcomes, and compare results across the two policy domains (collective action versus environmental regulation) to assess whether “rugged individualism” is more strongly related to opposition toward top-down regulation than to bottom-up action (hypothesis 2).

4.2 Step 2: Frontier exposure and political ideology

Next, we test the effect of TFE on contemporary Republican voteshare to replicate the relationship found in Bazzi et al. (2020) and to ensure we can use political ideology as a mediator. We estimate:

$$R_i = \alpha_0 + \alpha_1 TFE_i + G'_i \alpha_2 + H'_i \alpha_3 + \delta_{s(i)} + u_i. \quad (2)$$

Because centering is a linear transformation, equation (2) captures the same slope relationship as regressing R_i on TFE_i ; it only shifts the intercept. The coefficient α_1 therefore measures whether counties with higher frontier exposure tend to have higher contemporary Republican vote share. Observing significant results here does not only show consistency with the findings documented by Bazzi et al. (2020), but it also allows us to use R_i as a mediator in our descriptive mediation analysis in step 3a. Results for this regression can be found in the Appendix 7.

4.3 Step 3: Diagnostic Mediation-style decomposition using Republican vote share

A central question is whether the association between frontier exposure and climate attitude, equation (1), reflects frontier culture itself or whether it operates through contemporary political ideology as a channel. To explore this, we include Republican vote share in our baseline regression:

$$y_i = \gamma_0 + \gamma_1 TFE_i + \gamma_2 R_i + G'_i \gamma_3 + H'_i \gamma_4 + \delta_{s(i)} + \eta_i. \quad (3)$$

Comparing β_1 from (1) with γ_1 from (3) is a descriptive mediation decomposition: it shows how much of the baseline TFE coefficient is absorbed once we condition on county vote share.

4.4 Step 4: Moderation analysis using an partisanship interaction term

In addition to the decomposition above, we test whether the association between frontier exposure and climate attitudes varies systematically with Republican vote share: Is frontier exposure more strongly related to climate-policy opposition in more Republican-leaning counties (hypothesis 3)? Specifically, we estimate a moderation specification that allows the marginal effect of frontier exposure to depend on county Republican vote share by including an interaction between TFE_i

and R_i :

$$y_i = \mu_0 + \mu_1 TFE_i + \mu_2 R_i + \mu_3 (TFE_i \times R_i) + G'_i \mu_4 + H'_i \mu_5 + \delta_{s(i)} + \xi_i. \quad (4)$$

With mean-centering, μ_1 , the main effect of TFE_i , captures the relationship between frontier exposure and the outcome at national average Republican vote share, while the interaction term μ_3 indicates how this relationship changes as counties become more Republican relative to the national mean. The marginal effect of TFE implied by (4) is therefore: $\frac{\partial y_i}{\partial TFE_i} = \mu_1 + \mu_3 R_i$.

4.5 Identification of causal effects

The inclusion of historical and geographic controls as well as state fixed effects in our models reduces the effect of observable and state-level unobservable determinants. Additionally, clustering standard errors reduces the effect of unobserved regional determinants of counties with close proximity.

However, we do not claim full causal identification of the effect of frontier exposure on climate attitudes. We emphasize that equation (3) and (4) should *not* be interpreted as identifying a causal mediation channel through R_i . The YCOM county-level opinion estimates are generated using multilevel modeling and poststratification, and the estimation procedure includes electoral vote-share covariates to improve predictive accuracy. Conditioning on Republican vote share can therefore induce a partly mechanical relationship between R_i and the dependent variable y_i . Coefficients on R_i may be inflated and standard errors may be optimistic. We therefore interpret the mediation and moderation models as diagnostic tools rather than causal mechanism analysis. Accordingly, we do not interpret γ_2 as the causal effect of Republican vote share on climate attitudes, nor do we interpret the change from β_1 to γ_1 as the portion of the TFE effect attributable to a Republican-vote-share mediation channel. Remaining threats to causality include county-level unobservable variables correlated with both TFE_i and climate attitudes, and the fact that YCOM outcomes are generated estimates rather than directly observed population shares.

5 Results

5.1 Environmental Regulation

The results of the baseline regression, equation (1), for the effect of TFE on climate-policy opposition are moderate, yet positive and statistically significant. For our first survey question, the estimated percentage who do not say global warming should be a high priority for the next president and Congress in Table 4, the TFE coefficient is 0.279 but only statistically significant at the 10% level. For our second survey question, the estimated percentage who oppose requiring fossil fuel companies to pay a carbon tax in Table 5, it increases to 0.288 and becomes statistically significant at the 5% level. These estimates imply that a 10-year increase in frontier exposure is associated with roughly 0.28 to 0.29 p.p. higher opposition to top-down climate policy. Given a maximum TFE of 63 years and a minimum of one year, the implied difference between lowest- and highest-TFE counties is about 1.8 p.p.

Table 4: Estimated percentage who do not say global warming should be a high priority for the next president and Congress

	Baseline		Mediation		Moderation	
	β	se	β	se	β	se
TFE	0.279*	(0.143)	0.038	(0.062)	0.040	(0.062)
RepShare			33.855***	(0.599)	33.839***	(0.593)
TFE $\times$ RepShare					0.796**	(0.337)
Observations	2036		2036		2036	
R-squared	0.498		0.833		0.834	

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

All models include geographic and historical controls, as well as state fixed effects.

All models control for log 2020 population density.

Standard errors are clustered at 60km grid cell. All variables are centered.

Table 5: Estimated percentage who oppose requiring fossil fuel companies to pay a carbon tax and use the money to reduce other taxes (such as income tax) by an equal amount

	Baseline		Mediation		Moderation	
	β	se	β	se	β	se
TFE	0.288**	(0.138)	0.033	(0.041)	0.034	(0.040)
RepShare			35.868***	(0.546)	35.856***	(0.542)
TFE $\times$ RepShare					0.572**	(0.281)
Observations	2036		2036		2036	
R-squared	0.552		0.919		0.920	

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

All models include geographic and historical controls, as well as state fixed effects.

All models control for log 2020 population density.

Standard errors are clustered at 60km grid cell. All variables are centered.

The mediation column of each table includes county Republican vote share in the regression. In both top-down outcomes, the TFE coefficient becomes statistically insignificant from zero (between 0.033 and 0.038). At the same time, the coefficient on Republican vote share is large and highly significant (around 34–36 percentage points)¹¹.

The interaction term is positive and statistically significant at the 5% level in the moderation column indicating that the marginal association between TFE and climate-policy opposition becomes stronger as Republican vote share increases above its sample mean (which is around 65%).

5.2 Collective Action

Tables 6 and 7 summarize our findings, for the relationship between TFE and the collective action policy approach, which is positive and statistically significant at the 1% level for both outcomes. For the estimated percentage who think their local officials should do less to address global warming and the estimated percentage who think citizens should do less to address global warming, the point estimate is 0.171 and 0.172, respectively. This implies that a 10-year increase in frontier exposure is associated with roughly 0.17 p.p. higher opposition to bottom-up climate action, conditional on the controls and state fixed effects.

Table 6: Estimated percentage who think their local officials should do less to address global warming

	Baseline		Mediation		Moderation	
	β	se	β	se	β	se
TFE	0.171***	(0.064)	0.061**	(0.027)	0.062**	(0.027)
RepShare			15.471***	(0.294)	15.464***	(0.296)
TFE $\times$ RepShare					0.344**	(0.141)
Observations	2036		2036		2036	
R-squared	0.507		0.851		0.851	

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

All models include geographic and historical controls, as well as state fixed effects.

All models control for log 2020 population density.

Standard errors are clustered at 60km grid cell. All variables are centered.

As with the top-down outcomes, adding Republican vote share in our mediation regression reduces the estimated TFE coefficient. For the estimated percentage who think their local officials should do less to address global warming, the TFE coefficient becomes essentially zero (0.061), which we also observed for the top-down results. However, the coefficients become statistically significant at the 5% level (Table 6), which we did not find for the environmental regulation. For the estimated percentage who think citizens should do less to address global warming, it falls to 0.083 and stays highly significant at the 1% level (Table 7). Republican vote share remains very strongly associated

¹¹We want to emphasize the key limitation discussed earlier. YCOM outcomes are model-based estimates that use vote-share covariates in their construction, thus the coefficient appear to be overly optimistic and the R^2 increases sharply.

Table 7: Estimated percentage who think citizens should do less to address global warming

	Baseline		Mediation		Moderation	
	β	se	β	se	β	se
TFE	0.172***	(0.052)	0.083***	(0.026)	0.084***	(0.026)
RepShare			12.511***	(0.279)	12.505***	(0.282)
TFE $\times$ RepShare					0.299**	(0.141)
Observations	2036		2036		2036	
R-squared	0.543		0.829		0.829	

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

All models include geographic and historical controls, as well as state fixed effects.

All models control for log 2020 population density.

Standard errors are clustered at 60km grid cell. All variables are centered.

with both outcomes, and R^2 rises sharply ¹².

Tables 6 and 7 show positive and statistically significant interaction terms at the 5% level for both bottom-up outcomes. This implies that the association between frontier exposure and opposition to bottom-up action is stronger in more Republican counties (relative to the mean), consistent with the results from the top-down approach.

5.3 Meaning for Hypotheses

As we already mentioned in section 4.5, one should be careful about the interpretation of the results. We strongly recommend to not understand the numerical outcomes as causal relationships but rather as diagnostic tools. The most important aspect to consider is that the mediation analysis is generally not that useful¹³ and the effect of TFE mostly gets insignificant. We therefore choose to use the baseline regression to test hypotheses 1 and 2 and the moderation analysis to test hypothesis 3.

5.3.1 Hypothesis 1

The first hypothesis is generally true, higher TFE leads to stronger opposition towards climate policies. This can be seen in the four coefficients β in the baseline regressions, that are positive and significant on the 10%-confidence interval at least.

5.3.2 Hypothesis 2

The second hypothesis is also true in the baseline regression: when comparing the coefficient across the two policy implementation strategies, the effect of TFE is greater on climate policies that are considered top-down regulations than on climate policies that are more likely to be associated with

¹²These patterns again point to a tight empirical connection between partisanship and climate-policy opposition and should be treated as descriptive because vote-share covariates enter the YCOM estimation procedure.

¹³See section 4.5

bottom-up collective action. This can be seen by comparing the β 's from tables 4 and 5 with the β 's from tables 6 and 7. Fromer, representing the effect on TFE on opposition towards top-down regulation are generally larger than the latter, who represent the effect of TFE on bottom-up collective action¹⁴.

5.3.3 Hypothesis 3

The results of the moderation analysis also suggest that our third hypothesis is true as the β 's of the interaction terms are generally positive and significant on the 5%-confidence interval. This means, as already mentioned above, that counties with a higher republican vote share also experience a stronger effect of the TFE, i.e. the effect of the TFE gets stronger with a higher republican voting share.

6 Discussion

6.1 Implications

Our results have suggestive implications for US-climate politics. We not only find that the experiences of frontier societies and the culmination of a culture of "rugged individualism" leads to opposition towards climate policies¹⁵, but we also found evidence about how exactly this opposition is structured and preserved. The two main points that one could make using our analysis are the following: First, counties with high TFE dislike top-down regulation more than bottom-up collective action. This highlights the importance of climate policy design when dealing with ideological opposition towards coordinated problem-solving. Secondly and probably even more interesting is the fact that the effect of the TFE is greater in republican counties. That is how we find empirical evidence for the "stickiness" of "rugged individualism" in conservative areas. This could help explain why cultural habits that already match a community's political ideology tend to persist. In this way, political ideologies can reinforce themselves.

6.2 Validity and limitations

Now that we have interpreted our empirical findings and drawn political implications from them, it is important to keep in mind that there are still threats to validity that should be discussed.

A key internal validity challenge stems from the outcome variables. YCOM county-level measures are not directly observed but come from multilevel regression and poststratification (MRP), adding model-based noise beyond sampling error that clouds inference. Baseline relationships are precise and robust to controls and fixed effects, but the reported standard errors miss the extra uncertainty

¹⁴One could argue that the regression coefficients are not really comparable as we did not standardize the dependent variables, i.e. the YCOM questions. We account for this in the appendix and the results break down. Out of reasons mentioned in there, we still prefer to stick to our "main" results.

¹⁵Bazzi et al. (2020) already found a effect of TFE on the opposition towards CO₂-regulations. Our findings are generally consistent with their research. We replicate their results in the Appendix.

from the estimation step, so the results appear more precise than they really are. The MRP also uses geographic covariates like Democratic vote share for accuracy, creating mechanical ties between vote shares and YCOM climate attitudes, so specifications with Republican vote share are not causal. These concerns hit mediation analysis hardest. Since vote share shapes YCOM estimates, adding Republican vote share as a regressor mechanically boosts its coefficient and shrinks TFE’s effect. We thus treat mediation as a diagnostic decomposition, not causal proof.

Another limitation of the YCOM data is that county-level estimates are less precise than those at larger geographic scales. Howe et al. (2015) report a margin of error of ± 8 percentage points at the county level, compared to ± 5 at the state level. This is partly because a large share of counties (about 41%) have no actual survey respondents in the underlying data at all, meaning the model relies entirely on demographic and geographic formulas to generate those estimates rather than real local responses. As a result county-level estimates should be interpreted with caution.

Our outcomes are based on a few specific questions so they capture only a narrow slice of respondents overall climate attitudes. Categories like “top-down” or “bottom-up” can oversimplify attitudes, respondents may read items differently, and focusing on only two policies can miss broader views on regulation or collective action. Future research could add more YCOM questions as robustness checks. Also negatively worded items could add response bias from phrasing effects. Plus, cross-sectional data means we can not track attitude stability or how frontier effects evolve over time. Furthermore, climate attitude data are drawn from 2020, coinciding with the COVID-19 pandemic, which may have temporarily influenced public opinion on climate change in ways that could affect the present findings. Future research should examine whether the associations reported here replicate across multiple YCOM survey years to establish the temporal stability of the results. The analysis sticks to the U.S. and frontier settlement, so links to modern climate views may not travel to places with different histories, institutions, or cultures.

These limitations do not invalidate the analysis, but they shape how the results should be read. Future research should treat our results as indications of historical-cultural persistence in climate skepticism but only in a suggestive sense rather than as definitive proof. One has to keep in mind that survey data on climate attitudes at the county level is scarce. Most surveys are national because local sampling is costly, and no widely available dataset provides responses for all U.S. counties. Given these constraints, our approach using YCOM represents a feasible strategy with currently available data, and thus a first step in assessing whether the proposed historical mechanism is empirically plausible.

Future work could build on this by addressing the remaining identification limits. One promising direction would be to use county level survey data whenever such data become available, or to rely on alternative opinion measures that do not incorporate electoral covariates. Indeed, Howe et al. (2015) note that the partisan vote-share term in the MRP model can be dropped with substantively similar results. Re-estimating climate-opinion measures without this covariate, would allow researchers to probe the proposed mechanisms and move from suggestive descriptive evidence toward clearer causal inference.

7 Conclusion

This research examines whether the historical legacy of the American westward expansion continues to shape modern attitudes toward climate policy through the lens of “rugged individualism.” By linking historical Total Frontier Experience (TFE) with contemporary county-level data from the Yale Climate Opinion Map (YCOM), the study shows that frontier exposure remains a significant predictor of climate-policy opposition today, more than a century after the end of frontier settlement.

The empirical analysis supports three main conclusions. First, higher TFE is associated with stronger opposition to climate policies in general, consistent with the persistence of a culture emphasizing self-reliance and skepticism toward collective problem-solving. Second, the relationship is systematically stronger for top-down environmental regulation than for bottom-up collective action, indicating that frontier culture is particularly incompatible with centralized government intervention. Third, the moderation analysis shows that these relationships are stronger in counties with higher Republican vote shares, suggesting that frontier-linked cultural traits appear more persistent in conservative political environments.

These findings imply that the frontier legacy does not only predict whether climate policies are opposed, but also how that opposition is structured. Counties with greater frontier exposure exhibit relatively stronger resistance to centralized regulation than to bottom-up approaches, and this pattern is most pronounced in Republican-leaning areas. This supports the interpretation that historically rooted individualism continues to interact with contemporary political ideology in shaping climate-policy preferences. These results highlight the importance of climate policy design in regions shaped by frontier culture. Because high-TFE counties display a particularly strong aversion to centralized authority, policies that combine national targets with decentralized implementation may face less resistance than purely top-down regulatory approaches. Such institutional designs may align more closely with frontier-rooted preferences for autonomy and local decision-making while still enabling coordinated climate action.

At the same time, the findings must be interpreted with caution. The analysis relies on modeled county-level climate-opinion estimates derived from multilevel regression and poststratification, and full causal identification is not possible in this setting. The mediation and moderation results involving vote share are therefore best understood as descriptive rather than causal. Despite these limitations, the paper provides robust evidence of a persistent association between historical frontier exposure and contemporary climate-policy attitudes, contributing to the literature by linking long-run cultural persistence to modern environmental policy preferences.

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Appendix

Replication: TFE and Republican vote share

We reproduce the positive relationship between frontier exposure and contemporary Republican vote share highlighted by Bazzi et al. (2020), shown in Table 8. In our data, TFE is measured in decades. The estimated coefficient on centered TFE is 0.018 and highly significant, implying that an additional decade of frontier experience is associated with a 1.8 percentage points higher Republican vote share (since vote share is scale 0-1), conditional on controls and state fixed effects. This magnitude is similar to Bazzi et al. (2020), who reported a 2 p.p. increase in Republican vote share per decade of TFE. The 2 p.p. increase is relative to the mean over the five presidential elections from 2000 to 2016. We use the 2020 presidential election which could explain the 0.2 p.p. difference.

Table 8: The Effect of Frontier Experience on Republican Vote Share (2020)

	β	se
TFE	0.018***	(0.003)
Observations	2036	
R-squared	0.513	

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Regression includes geographic and historical controls, as well as state fixed effects.

All models control for log 2020 population density.

Standard errors are clustered at 60km grid cell. The Dependent variable is Republican vote share.

Z-standardized outcomes (robustness for cross-question comparisons)

Our main results are reported in percentage points, which keeps the interpretation intuitive and directly comparable to the underlying survey quantities. However, the four YCOM outcomes differ in their baseline dispersion across questions. As a robustness check, we z-standardizing each outcome, $y_i^z = (y_i - \bar{y})/sd(y)$. This transformation expresses coefficients in standard-deviation units of the outcome and facilitates comparisons of effect magnitudes across questions.

Relative to the main results, baseline effect sizes are therefore smaller in absolute numeric terms, with TFE associated with roughly 0.06–0.09 sd higher climate-policy opposition per additional decade of frontier exposure. In the main text, we compared coefficients in percentage points and found larger absolute effects for the two top-down outcomes than for the bottom-up collective-action outcomes, which we interpreted as support for Hypothesis 2. In standardized units, this ordering is not robust: the baseline TFE coefficients are slightly larger for the bottom-up outcomes (Tables 11 and 12) than for the top-down outcomes (Tables 9 and 10). This difference arises because the survey questions differ in their cross-county dispersion. The coefficients reflect the effect relative to typical variation in that specific question rather than an absolute change in percentage points. We therefore conclude that Hypothesis 2 is supported in absolute (p.p.) terms,

Table 9: Estimated percentage who do not say global warming should be a high priority for the next president and Congress (standardized)

	Baseline		Mediation		Moderation	
	β	se	β	se	β	se
TFE	0.056*	(0.028)	0.008	(0.012)	0.008	(0.012)
RepShare			6.735***	(0.119)	6.731***	(0.118)
TFE $\times$ RepShare					0.158**	(0.067)
Observations	2036		2036		2036	
R-squared	0.498		0.833		0.834	

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

All models include geographic and historical controls, as well as state fixed effects.

All models control for log 2020 population density.

Standard errors are clustered at 60km grid cell. All variables are centered; outcome is z-standardized.

Table 10: Estimated percentage who oppose requiring fossil fuel companies to pay a carbon tax and use the money to reduce other taxes (such as income tax) by an equal amount (standardized)

	Baseline		Mediation		Moderation	
	β	se	β	se	β	se
TFE	0.057**	(0.027)	0.006	(0.008)	0.007	(0.008)
RepShare			7.048***	(0.107)	7.045***	(0.106)
TFE $\times$ RepShare					0.112**	(0.055)
Observations	2036		2036		2036	
R-squared	0.552		0.919		0.920	

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

All models include geographic and historical controls, as well as state fixed effects.

All models control for log 2020 population density.

Standard errors are clustered at 60km grid cell. All variables are centered; outcome is z-standardized.

but it is not robust in standardized effect terms. The mediation and moderation patterns mirror the main text. Adding Republican vote share reduces the TFE coefficient. They become close to zero for the top-down outcomes and are reduced by about 50% for the bottom-up outcomes. The Republican vote share itself remains large and highly significant, and R^2 increases sharply. As discussed in the empirical strategy, these mediation results should be interpreted descriptively because vote-share covariates enter the YCOM estimation procedure. Finally, the interaction between TFE and Republican vote share remains positive and statistically significant across all outcomes in the moderation specifications. Expressed in sd units, these interaction effects are also more comparable across questions, reinforcing Hypothesis 3 that the association between frontier exposure and climate-policy opposition is stronger in more Republican counties.

Table 11: Estimated percentage who think their local officials should do less to address global warming (standardized)

	Baseline		Mediation		Moderation	
	β	se	β	se	β	se
TFE	0.075***	(0.028)	0.027**	(0.012)	0.027**	(0.012)
RepShare			6.814***	(0.129)	6.811***	(0.130)
TFE $\times$ RepShare					0.151**	(0.062)
Observations	2036		2036		2036	
R-squared	0.507		0.851		0.851	

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

All models include geographic and historical controls, as well as state fixed effects.

All models control for log 2020 population density.

Standard errors are clustered at 60km grid cell. All variables are centered; outcome is z-standardized.

Table 12: Estimated percentage who think citizens should do less to address global warming (standardized)

	Baseline		Mediation		Moderation	
	β	se	β	se	β	se
TFE	0.085***	(0.026)	0.041***	(0.013)	0.042***	(0.013)
RepShare			6.211***	(0.138)	6.208***	(0.140)
TFE $\times$ RepShare					0.148**	(0.070)
Observations	2036		2036		2036	
R-squared	0.543		0.829		0.829	

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

All models include geographic and historical controls, as well as state fixed effects.

All models control for log 2020 population density.

Standard errors are clustered at 60km grid cell. All variables are centered; outcome is z-standardized.

While standardizing the dependent variables allows for comparison across outcomes, we keep the percentage-point comparison in the main text for the following reasons. First, Climate attitudes taking from identical survey response scales, allowing different coefficients to be interpreted as

changes in expressed policy oppositon rather than relative variation in dispersion. Effects measured in percentage-points are more directly interpretable in regards to our theoretical argument, which concerns the absolute shifts in policy preferences associated with frontier exposure. Second, standardizing our outcome variable removes variances across survey questions. This could get rid of potentially important differences in opinion polarization between policy domains. Overall, our analysis suggests that cross-outcome results are sensitive to rescaling assumptions rather than underlying relationships. For transparency and interpretability, we report standardized results as robustness checks but base our main analysis on the original survey metric.

Who did what?

Hannah Arndt: Subsection 3.1 (Dependent Variable with table), Subsection 3.3 (Controls), Subsection 6.5 (Validity and Limitations), data research and data collection (Population Density, Socioeconomic Controls and extreme weather (it was decided not to include the latter two)), data preparation (cleaning, converting and merging Population Density, Socioeconomic Controls and Extreme weather (it was decided not to include the latter two)), adjusted the analysis to account for Population density as a Control, data visualization

Bastian Reßing: Section 1 (Introduction), Section 2 (Background), Subsection 5.4 (Meaning for Hypothesis), Subsection 6.1 (Implications), setting up LaTeX-environments, setting up layout for paper and presentation, data research, literature review, adding sources, adding pictures of maps

Saskia Schatzmayr: Section 4 (Empirical Strategy), Section 5 (Results) [except subsection 5.4], Appendix (Replication: TFE and vote share, Robustness check), data research, data collection (YCOM data, TFE and historical + geographic controls from Bazzi et al. (2020)'s replication file), replicating Bazzi et al. (2020) findings, data preparation (cleaning and merging TFE, YCOM and vote share data), planning empirical strategy, data analysis, data visualization (all graphs), code cleaning and ensuring reproducibility

Margaret Vincent: Section 3 Overview (Data), Subsection 3.2 (TFE with the table), Subsection 3.4 (Voting Data with the table), Section 7 (Conclusion), one paragraph in Subsection 3.3 (Controls), data research and collection (Republican Vote Share Data)